

Auteur: Noran Ben Said
Studierichting: Informatica
Oefeningenreeks 4A nr. 19.1

$$I = \int e^x \sin x dx$$

$$\text{Stel } \begin{cases} u = e^x \\ dv = \sin x dx \end{cases} \Rightarrow \begin{cases} du = e^x dx \\ v = -\cos x \end{cases}$$

$$\Rightarrow I = -e^x \cos x + \int e^x \cos x dx$$

$$\text{Stel } \begin{cases} u = e^x \\ dv = \cos x dx \end{cases} \Rightarrow \begin{cases} du = e^x dx \\ v = \sin x \end{cases}$$

$$\Rightarrow I = -e^x \cos x + e^x \sin x - \int e^x \sin x dx$$

$$\Rightarrow 2I = -e^x \cos x + e^x \sin x + C$$

$$\Rightarrow I = \frac{e^x (\sin x - \cos x)}{2} + C$$

References